

UNIVERSITY OF CAPE TOWN

SCHOOL OF ECONOMICS

ECO4104Z

Economic Analysis of Financial Markets

FINAL EXAM

April 2024

Time: 2 hours

Total: 100 marks

**INSTRUCTIONS:**

**ALL QUESTIONS ARE COMPULSORY. THE VALUE OF EACH QUESTION AND  
SUB-QUESTIONS IS INDICATED IN PARENTHESES.**

1i) 1.

2.

ii) Endog vars. are correlated with the error term  $\epsilon$  are determined within a model vs  
exog. vars. that are uncorrelated  $\epsilon$ , determined outside the model

2.

3i)

**Question 1: [8 marks]**

Endogeneity issues are a major problem in most econometric research analysis. Use the following equation to answer the question below:

$$y_t = \alpha_0 + \alpha_1 x_t + \alpha_2 z_t + \varepsilon_t$$

Where:  $y_t$  is the dependent variable.  $x_t$  and  $z_t$  are independent variables, and  $\varepsilon_t$  is the disturbance term.

- (i) Using the econometric meaning, explain two cases where a research analyst is likely to experience the problem of endogeneity using the above equation. (5 marks)
- (ii) Using economic meaning, explain the difference between exogenous and endogenous variables. (3 marks)

**Question 2: [7 marks]**

Dickey and Fuller (1979) consider three different types of regressions:

$$\Delta y_t = \gamma y_{t-1} + \sum_{i=2}^p \beta_i \Delta y_{t-i+1} + \varepsilon_t$$

$$\Delta y_t = a_0 + \gamma y_{t-1} + \sum_{i=2}^p \beta_i \Delta y_{t-i+1} + \varepsilon_t$$

$$\Delta y_t = a_0 + \gamma y_{t-1} + a_2 t + \sum_{i=2}^p \beta_i \Delta y_{t-i+1} + \varepsilon_t$$

Describe the differences between these models. After identifying the parameter of interest, briefly explain how you would use the test statistic to determine whether a unit root is present in the data series. (7 marks)

**Question 3: [6 marks]**

Time series data often exhibits trend sequences.

- (i) Explain two different types of trends encountered by econometricians. (4 marks)
- (ii) Explain how you can eliminate these trends. (2 marks)

**Question 4: [8 marks]**

You are given the following equations:

$$(i) \quad y_t = y_{t-1} + \varepsilon_t$$

$$(ii) \quad y_t = y_{t-1} + t + \varepsilon_t$$

How can you make the models in (i) and (ii) stationary? (5 marks)

(iii) Money supply data set for country Y can be represented by the following equation:

$y_t = y_{t-1} + \varepsilon_t$ . You are further informed that the money supply has a seasonal unit root. Suppose you observe monthly frequency for money supply, how can you make this model stationary? (3 marks)

### Question 5: [6 marks]

Consider the first-order autoregressive process in modelling volatility

$$x_t = a_0 + a_1 x_{t-1} + \varepsilon_t \quad \text{where } |a_1| < 1$$

(i) Find the conditional mean of  $x_{t+1}$

(ii) Find the conditional mean of  $x_{t+2}$

(iii) Find the unconditional mean of  $x_{t+1}$  (6 marks)

### Question 6: [14 marks]

Jabeen and Khan (2014) estimated a GARCH (1,1) model analysing exchange rate volatility by Macroeconomic fundamentals in Pakistan. An extract of their variance equation is provided below:

|              | Pakistan Rupee against US\$ |        | Pakistan Rupee vs Japanese Yen |        |
|--------------|-----------------------------|--------|--------------------------------|--------|
| Variable     | coefficient                 | Prob.  | coefficient                    | Prob   |
| constant     | 0.0589                      | 0.5920 | 0.0000                         | 1.0000 |
| Resid(-1)^2  | 0.1545                      | 0.0000 | 0.0596                         | 0.1658 |
| GARCH(-1)    | 0.5108                      | 0.0000 | 0.7806                         | 0.0000 |
| LM ARCH-test | 0.4011                      | 0.9458 | 1.0950                         | 0.3652 |

(i) Are there GARCH (1,1) effects for both the Pakistan Rupee vs the US dollar and the Japanese Yen? (8 marks)

(ii) One of the diagnostic tests they did is the LM ARCH test. Using the information in the Table above, comment what the LM ARCH-test is informing us here. (6 marks)

### Question 7: [17 marks]

i) What is the **reduced form of a VAR process**? Why is it important to derive it? (5 marks)

ii) What type of a **problem could one encounter when estimating an unrestricted VAR process**? Explain. (5 marks).

iii) A researcher estimates a VAR model with six variables. The optimal lag length for this process is four based on AIC criteria. Based on this information, (a) how many coefficients will this model estimate? (2 marks) (b) how will you use the t-tests calculated to judge the adequacy of the model? (5 marks)

### Question 8: [6 marks]

Stock & Watson (1988) propose the following decomposition for I(1) variables:

$$y_t = \mu_{yt} + \varepsilon_{yt},$$

$$z_t = \mu_{zt} + \varepsilon_{zt}; (i = y, z)$$

Where  $\mu_{it}$  is the random walk component and  $\varepsilon_{it}$  is the stationary component. Find the linear combination of these variables and determine the conditions under which a multivariate expression of the variables may be stationary. (6 marks)

### Question 9: [10 marks]

A good way to examine the relationship between cointegration and error correction is to study the properties of the simple VAR model. Considering the following standard form VAR, use matrix algebra to find the solution for  $y_t$  and  $z_t$ . (10 marks)

$$y_t = a_{11}y_{t-1} + a_{12}z_{t-1} + \varepsilon_{yt}$$

$$z_t = a_{21}y_{t-1} + a_{22}z_{t-1} + \varepsilon_{zt}$$

### Question 10: [9 marks]

The error correction form of a model may be expressed as:

$$\Delta x_t = \pi_0 + \pi x_{t-1} + \pi_1 \Delta x_{t-1} + \pi_2 \Delta x_{t-2} + \dots + \pi_p \Delta x_{t-p} + \varepsilon_t$$

(i) if all  $\pi$  are zero, what does it imply? (2 marks)

(ii) if one of  $\pi$  is not zero, what does it imply? (2 marks)

(iii) Explain why the Engle-Granger cointegration technique is inappropriate in the VAR context. Be precise (5 marks)

### Question 11: [9 marks]

(i) Describe how you would carry out the ARDL cointegration technique (6 marks).

(ii) What are the advantages of the ARDL technique over the Johansen technique and the Engle-Granger technique. (3 marks)

